

INTRODUCTION TO ENGINEERING • LECTURE 3

Systems Thinking II

Process Flow Diagrams and How Perspective Shapes What We See

Dr. Mohammad Reza Bahrami

Faculty of Energy Engineering — Tashkent University of Technology

Disclaimer: this lecture references material adapted from Kosky, Balmer, Keat & Wise, Exploring Engineering (5th ed.), and from the instructor's own course slides, with modifications. No authorship of third-party material is claimed.

RECAP

Last Week, in One Breath

System — interacting elements and processes that turn inputs into outputs.

Boundary — a line the engineer chooses — a choice, not a fact about the world.

Operating context — what constrains the system from just outside its boundary.

Stakeholders — who uses it, runs it, or is affected by it.

Today — what happens inside the boundary — and who sees it how.

TODAY'S ROADMAP

Agenda

1 Opening the Box

From what a system contains to what it does

2 The Language of PFDs

Symbols, rules, and common mistakes

3 Worked Example: Dining Hall

Levels of detail, decisions, waits, and swimlanes

4 Different Perspectives

Why five stakeholders draw five diagrams

5 Wrap-Up

Key takeaways and independent study

Interactive moments

Fix the diagram · Three lenses on the bus route

WARM-UP · 3 MINUTES

Inside or Outside? The Dining Hall Again

Last week's open question. Show of hands, then defend your choice in one sentence.

The supplier's delivery truck

Inside the dining hall boundary, or outside?

The meal-plan billing office

Inside the dining hall boundary, or outside?

There is no single right answer, only a justified one.

PART 1

Opening the Box

From what a system contains to what a system does.

OPENING THE BOX

Elements Are Nouns, Processes Are Verbs

Elements

The things and people in the system.

Cooks · Ovens · Trays · Tables · Students

Processes

What those elements do, in order.

Receive · Store · Prepare · Serve · Pay · Clear

Example: a campus dining hall.

Lecture 2 listed the parts. A process flow diagram shows what they do together.

OPENING THE BOX

One System, Three Levels of Detail

LEVEL 0

Black box

Inputs go in, outputs come out. Nothing inside is shown.

Dining hall: food, staff, students in; meals, waste, payments out.

LEVEL 1

Main steps

The handful of major steps inside the boundary.

Receive → Store → Prepare → Serve → Pay → Clear.

LEVEL 2

Zoom in

One Level 1 step opened up into its own diagram.

Prepare meals: Wash → Chop → Cook → Hold hot.

Choose the level that fits your question, and keep one level per diagram.

PART 2

The Language of Process Flow Diagrams

A few symbols and a few rules make a diagram anyone can read.

What Is a Process Flow Diagram?

A **process flow diagram (PFD)** shows the steps of a process in order, and what flows between them.

Sequence

What happens first, next, and last.

Flow

What moves between steps: material, people, information, money, energy.

Problems

Gaps, waits, loops, and hand-offs become visible.

Kosky et al. use the same idea for the engineering design process itself (Fig. 21.1): the design process is also a process.

PROCESS FLOW DIAGRAMS

The Building Blocks



Start / End

Where the process begins
and finishes



Step

One action, named verb +
noun



Decision

A question with labeled
exits: Yes / No



Flow

Order of steps, and what
moves



Wait / queue

Where things sit and wait
(our course convention)

Five symbols are enough for this course. Consistency matters more than which symbols you choose.

PROCESS FLOW DIAGRAMS

Five Rules for a Readable Diagram

- 1 Start and end clearly.** One start, and every path reaches an end.
- 2 Name steps with a verb + noun.** “Receive deliveries”, not “Delivery”.
- 3 Label the arrows.** Say what moves: food, students, information, money.
- 4 Give every decision its exits.** Each Yes and each No must lead somewhere.
- 5 Keep one level of detail.** Arrows crossing the boundary are your inputs and outputs.

PROCESS FLOW DIAGRAMS

Every Arrow Carries Something

Material

Ingredients, trays, food waste

People

Students moving through the queue

Information

Menu, orders, meal-plan check

Money

Payments, supplier invoices

Energy

Gas, electricity, hot water

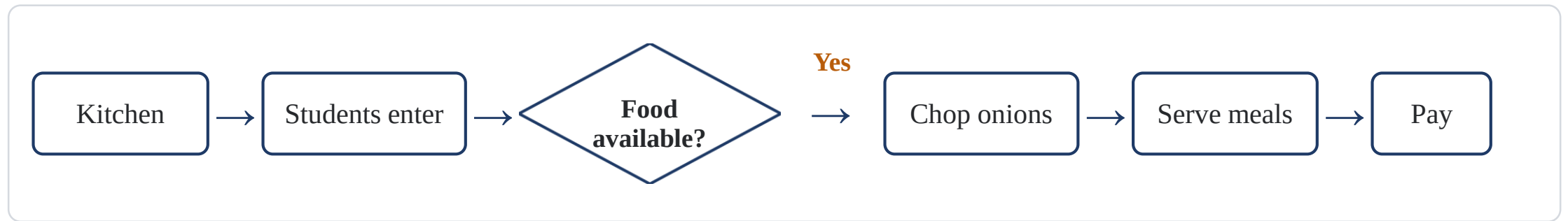
The same five categories we used for inputs and outputs in Lecture 2. Example: the dining hall.

Label each arrow with the thing that moves. It shows what the next step actually needs.

INTERACTIVE · 5 MINUTES

Fix the Diagram

This dining hall diagram breaks our rules. In pairs, find at least four errors.



Hints: starts and ends · labels · decisions · levels of detail · arrows

Fix the Diagram: Answer Key

#	Problem	Fix
1	No start or end symbols	Add Start and End
2	“Kitchen” is a place, not a step	Rename it: “Prepare meals”
3	The decision has only a Yes exit	Add a No exit, e.g. “Prepare more meals”
4	“Chop onions” is Level 2 detail	Keep one level: “Prepare meals”
5	Arrows carry no labels	Label what moves: students, meals, money

You may have found more. The rules matter because a reader should never have to guess.

PART 3

Worked Example: The Dining Hall

Building one diagram, one level at a time.

WORKED EXAMPLE

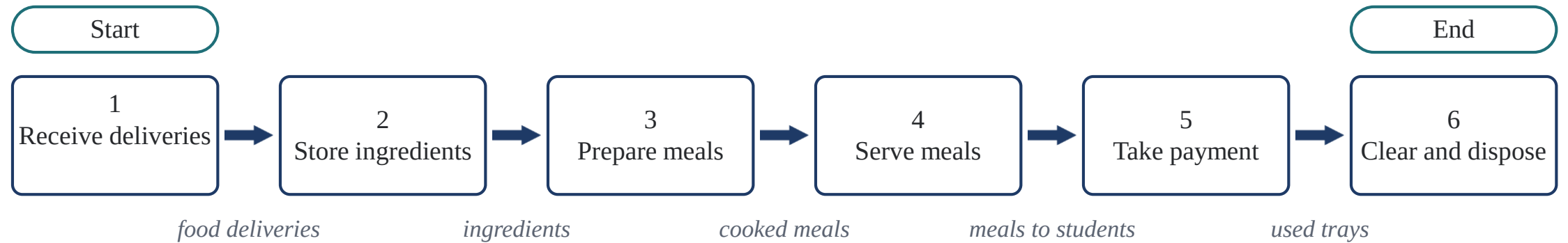
Level 0: The Black Box



Operating context: health code · opening hours · kitchen budget · building layout

WORKED EXAMPLE

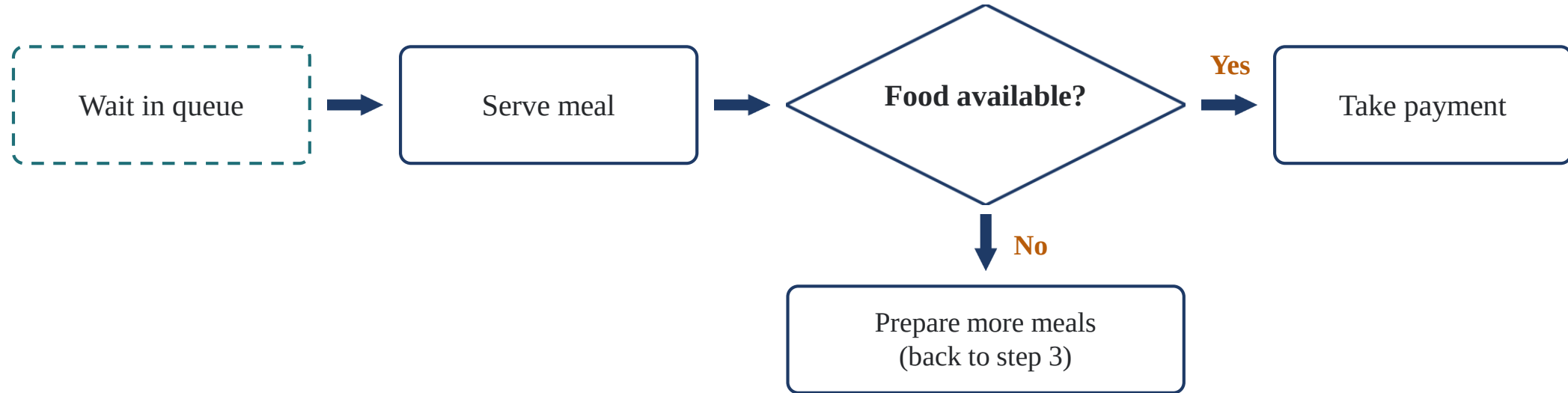
Level 1: The Main Steps



Compare with Level 0: deliveries enter at step 1 and students at step 4; payments leave at step 5 and waste at step 6.

WORKED EXAMPLE

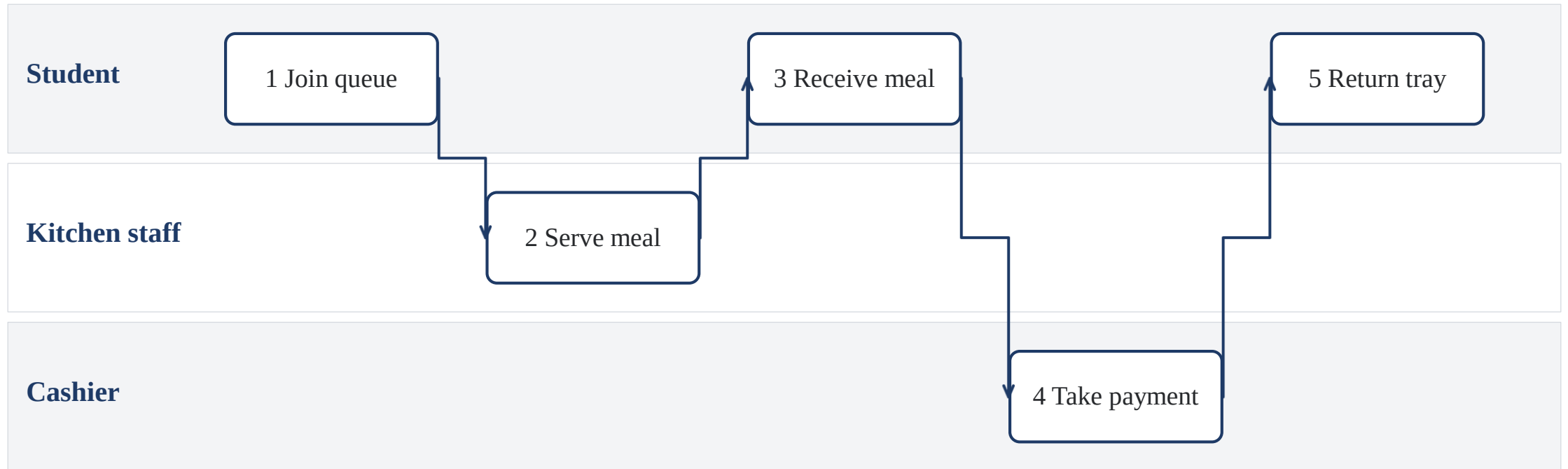
Adding Decisions and Waits



A dashed box marks a wait; a diamond marks a choice. Queues and decisions are where systems slow down. We measure them in Week 9.

WORKED EXAMPLE

Swimlanes: Who Does Each Step?



Numbers show the order. Every hand-off between lanes is a place where delays and errors can hide.

WORKED EXAMPLE

Back to the Boundary: Truck and Billing Office

The delivery truck

Usually **outside**. Deliveries cross the boundary as an input to step 1. The truck is a stakeholder that supplies the system, but none of our steps happens to the truck itself.

The meal-plan billing office

It depends where you draw the line. Outside if the diagram ends at “Take payment”; inside if it includes “Verify meal plan” and “Update balance”.

Either answer is defensible, if you state the boundary and keep it consistent.

Numbers on the Arrows



Nothing accumulates, so inflow equals outflow: $1000 - (250 + 200) = 550$ tons/day to landfill.

Source: Kosky et al., Exploring Engineering, Ch. 18, Example 18.2.

PART 4

Same System, Different Perspectives

Why five stakeholders draw five different diagrams.

PERSPECTIVES

A Diagram Is a Model With a Purpose

The London Underground map, designed by Harry Beck in the early 1930s, gives up real distances and street layout.

What it leaves out

Real distances · Street layout · Surface geography

What it keeps

Stations · Lines · Connections and interchanges

Every diagram keeps what serves its purpose and leaves out the rest. That is what makes a model useful. Weeks 4–5 return to this idea.

PERSPECTIVES

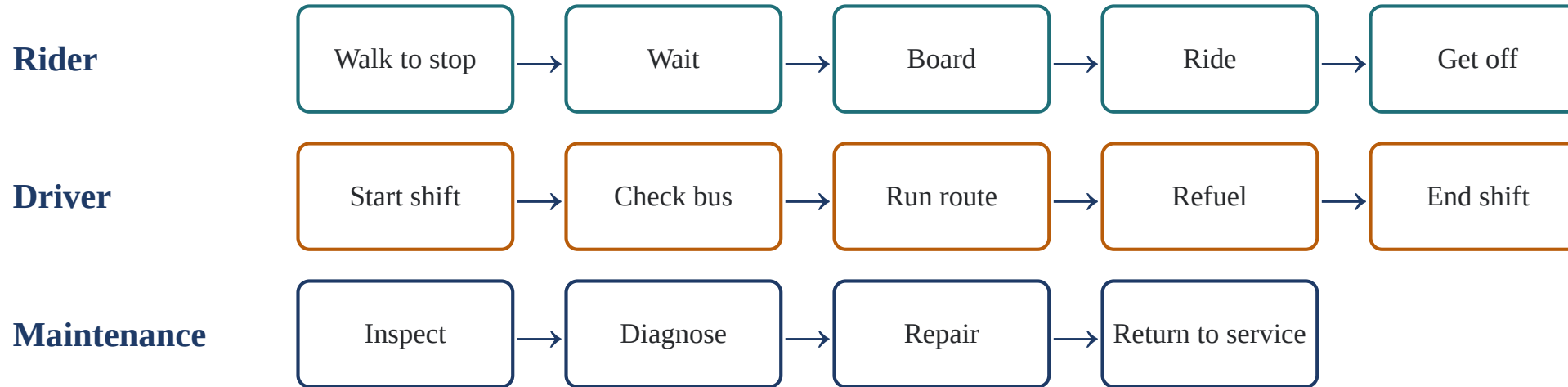
The Campus Bus Route Through Five Pairs of Eyes

Stakeholder	Their process	Their idea of success
Rider	Walk to stop → wait → board → ride → get off	Short wait, low fare, feels safe
Driver / dispatcher	Start shift → check bus → run route → refuel → end shift	On time, safe, manageable schedule
Maintenance crew	Inspect → diagnose → repair → return to service	Safe buses, planned downtime
University administration	Set budget → set timetable → review ridership → adjust	Cost per rider, coverage, few complaints
Neighbor	Hear bus pass → notice traffic → contact university	Quiet street, less traffic

Same system. Five different processes, and five different meanings of “good”.

PERSPECTIVES

Three of Those Diagrams, Side by Side



Different starting points, different steps, different end points, all describing one bus route.

PERSPECTIVES

What the Differences Tell Us

Different boundaries

The rider's picture starts at the stop; the driver's starts at the depot.

Different steps

Each person includes what they see, and leaves out what they don't.

Different success

Faster, safer, cheaper: Lecture 2's tension appears as competing definitions of good performance.

No diagram is wrong. Each is a model built for a purpose. Good engineers draw several, and ask whose view is missing.

INTERACTIVE · 10 MINUTES

Three Lenses on the Campus Bus Route

- 1 Each table receives one role.
- 2 In 5 minutes, sketch a 5-step flow of the bus route from that role's point of view. Label the arrows.
- 3 Post it, then compare: what is included, what is left out, and where do the diagrams conflict?

Rider

Driver / dispatcher

Maintenance crew

Administration

Neighbor

Independent Study: Apply It to Your System

- 1 Draw a Level 0 diagram of your system, then a Level 1 diagram with 5–8 steps.
- 2 Label every arrow, and mark where arrows cross your boundary.
- 3 Redraw one part from a second stakeholder's point of view.
- 4 Ask each stakeholder what “performing well” means to them.

Nothing to hand in. This is your working material, and it prepares the next step: turning observations into a well-defined problem.

WRAP-UP

Key Takeaways

Open the box in levels

Level 0 shows inputs and outputs; Level 1 the main steps; Level 2 zooms in.

Use a few symbols, consistently

Start/end, step, decision, arrow, wait. Verb + noun labels.

Every arrow carries something

Material, people, information, money, or energy.

Boundary and diagram must agree

Arrows crossing the boundary are inputs and outputs.

Diagrams are models with a purpose

They keep what serves the question and leave out the rest.

Perspective changes the picture

Stakeholders draw different diagrams and define success differently.